MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS NICKEL CHROMIUM ALLOY, TYPE I, CLASS 1, COMPOSITION B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Cr	Ni	Fe	C	Mn	Si	O_2
18.0 21.0	76.0 80.0	1.0 max.	0.25 max.	2.5 max.	1.5 max.	(1)

(1) Powder particles shall be essentially oxide-free.

Particle Size/Distribution

-230 Mesh (-63 µm)	95.0 min.
-270 Mesh (-53 µm)	80.0 min.

Sprayed Coating Properties

Bond Strength	31.0 MPa (4500 psi) min.
Hardness	Rockwell 15T, 90 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 3% max. Unmelted Particles - 10% max.

AMSC N/A FSC 3439

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Deposit Efficiency and Deposition Rate

70% min for N_2/H_2 @ 16.0 lb/hr
50% min for A/H_2 @ 16.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	B50TF40, Class B
Pratt & Whitney Aircraft	PWA 1317
Rolls-Royce, Ltd.	MSRR 9507/27

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/07)

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